UnB - FCTE
Faculdade de Ciências e Tecnologias em Engenharia

FGA0201: Digital Integrated Circuit Design (Projeto de Circuitos Integrados Digitais – PCID)

Semester: 2/2026
Requirements: Digital Systems 2 **or**
Digital Electronics Theory 2 **and** Digital Electronics Laboratory 2
Course Meets: Tuesdays/Thursdays, 8am -10am
Room: Lab. SS - UED
Instructor: Gilmar Beserra (gbeserra@unb.br)
Office Hours: Room 24 - UED (by schedule)
PCID 2/2026 group on WhatsApp: https://l1nk.dev/qrpkimf

Course Description

This course covers the basic concepts of CMOS-based digital integrated circuits by using CAD tools to the design, simulation, implementation, and verification of low and medium complexity digital circuits as case studies. The course topics are:

1. Introduction: digital vs analogic
2. MOS transistors
3. CMOS Logic
    a. Inverter
    b. Logic gates
    c. Pass transistor and transmission gates
    d. Tristates
    e. Multiplexer
    f. Sequential circuits
4. CMOS Fabrication
    a. Fabrication processes
    b. Design rules
    c. Layout
5. Design Flow
    a. Specification and modeling
    b. RTL design
    c. Logic synthesis
    d. Physical synthesis

By the end of the semester, the students will be able to:

- Define a MOS transistor and how it works
- Analyze digital gates circuits
- Describe how integrated circuits are fabricated

- Design and implement a digital circuit using all the steps and tools described on the design flow

Management System

The following platforms will be used during the course:

- **Moodle**: assignments and activities will be available here. In order to have access, the students need to register for the **FGA0201 - Projeto de Circuitos Integrados Digitais - 2/2026** course on **Aprender 3** (https://aprender3.unb.br/) using the code **FGA0201**.
- **Notion**: recorded classes, syllabus and all the course materials and information are available at the course website on Notion (free for students): https://l1nq.com/4f2c7l4 .

Grading and Assignments

The students will be evaluated based on the following activities:

- Laboratory assignments: 40%
- Exam (December,4th): 30%
- Final Project (December,9th): 30%

10 laboratory experiments will be performed during the semester and the students must submit a short report after each one.

- Lab. 1: MOS transistor simulation
- Lab. 2: Inverter characterization
- Lab. 3: Logic gates simulation
- Lab. 4: Logic gates layout
- Lab. 5: RTL design (combinational)
- Lab. 6: RTL design (sequential)
- Lab. 7: Logic Synthesis using Genus
- Lab. 8: DFT using Genus
- Lab. 9: ATPG using Modus
- Lab. 10: Physical Synthesis using Innovus

Grading Policy

The final grade will be calculated as follows:

$$\textit{Final} = 0.4*\textit{AVG(Lab. Assignments)} + 0.3*\textit{Exam} + 0.3*\textit{(Final Project)}$$

Each laboratory assignment will be graded from **0,0 to 10,0** and **AVG(Lab. Assignments)** means the average of all laboratory assignments. Late work will not be accepted.

Students will be divided into groups in order to implement a **Final Project** by the end of the semester. Each group must write a report about their project and submit it after the exam. To be approved, students must attend **at least 75%** of the classes and their **final grade should be equal or bigger than 5,0**.

In order to make the transmission of concepts and the improvement of the approach used possible, it is essential that the submitted assessments reflect only individual effort and understanding.

The use of AI tools, the submission of assessments copied in whole or in part (plagiarism), or the mere inclusion of a name without proper participation undermines the instructor's ability to evaluate the students' understanding, depriving them of the opportunity to master the content. Therefore, academic integrity is expected to be practiced.

If a breach of this integrity is found, the students in question will receive a grade of zero. In case of recurrence, they will fail the course, and the University will be notified for the applicable penalties.

Basic References

[1] Weste, N. H. E., Harris, D. M., **CMOS VLSI Design: A Circuit and Systems Perspective**, 4th ed. Addison-Wesley, 2011.

[2] Rabaey, Jan M., Chandrakasan, A., Nikolic, B., **Digital Integrated Circuits: A Design Perspective**, 2nd ed., Prentice Hall, 2003.

Complementary References

[1] Brunvand, E., **Digital VLSI Chip Design with Cadence and Synopsys CAD Tools**, Addison-Wesley, 2009.

[2] Uyemura, J. P., **Introduction to VLSI Circuits and Systems**, Wiley, 2001.

[3] Hodges, D., Jackson, H., **Analysis and Design of Digital Integrated Circuits**, McGraw-Hill, 2003.

[4] Kang, Sung-Mo.; Leblebici, Yusuf, **CMOS Digital Integrated Circuits Analysis & Design**, 3rd ed., McGraw-Hill, 2002.